SURVEYING LAB

Subject Code: 23A01301P

Lab Manual – 12 Field Works

B.Tech Civil Engineering | II Year CE – I Semester | JNTUA R23 | L-T-P-C: 0-0-3-1.5

Course Objectives

- Know various linear and angular measuring instruments; take linear and angular measurements.
- Determine area and volume from survey data.
- Know modern equipment such as total station; draft field notes from survey data.

List of Field Works

No.	Field work
1	Chain survey of road profile with offsets (road widening)
2	Distance between two inaccessible points using compass
3	Plane table survey – area of a boundary by radiation
4	Fly levelling – height of instrument method
5	Fly levelling – rise and fall method
6	Theodolite – horizontal and vertical angles by repetition
7	Theodolite – distance between two inaccessible points
8	Theodolite – height of a far object
9	Total station – area and perimeter
10	Total station – distance between two inaccessible points
11	Setting out a curve
12	Determining levels of contours

Note: Sample observations below are illustrative; record your own field readings in the tables and recompute.

Experiment 1: Chain Survey of Road Profile with Offsets (Road Widening)

Aim: To survey a road stretch by chaining and offsets and determine the area of widening required.

Apparatus: Metric chain/tape (20 m or 30 m), ranging rods, arrows, cross staff / optical square, pegs, plumb bob, field book.

Theory / Formulae

- Baseline is run along the road centre line; chainage is marked at fixed intervals (say 20 m).
- Perpendicular offsets are measured to the left and right edges at each chainage.
- Area between baseline and a boundary by trapezoidal rule: $A = (d/2)[O_0 + O_n + 2(O_1 + O_2 + \ldots + O_{n-1})]$
- Simpson's rule (even number of intervals): $A = (d/3)[O_0 + O_n + 4(O_1 + O_3 + \ldots) + 2(O_2 + O_4 + \ldots)]$
- Road width at a chainage = left offset + right offset; widening = proposed width − existing width.

Procedure

1. Fix the baseline along the road centre and ranging rods at both ends; chain along it.
2. Mark chainages at every 20 m with pegs/arrows.
3. At each chainage set the cross staff/optical square and measure the left and right offsets to the road edges.
4. Enter chainage and offsets in the field book (sketch with north line).
5. Compute widths, required widening and area by trapezoidal/Simpson's rule.

Observation Table

Chainage (m)	Left offset (m)	Right offset (m)	Width (m)	Proposed 10 m: widening (m)
0	3.2	3.0	6.2	3.8
20	3.4	3.1	6.5	3.5
40	3.0	3.3	6.3	3.7
60	3.5	3.2	6.7	3.3
80	3.1	3.0	6.1	3.9

Sample Calculation

Widening offsets: 3.8, 3.5, 3.7, 3.3, 3.9 at $d = 20$ m, $n = 4$ intervals.

Trapezoidal: $A = (20/2)[3.8 + 3.9 + 2(3.5 + 3.7 + 3.3)] = 10 \times 28.7 = 287\ m^2$

Simpson: $A = (20/3)[3.8 + 3.9 + 4(3.5 + 3.3) + 2(3.7)] = 6.667 \times 42.3 = 282\ m^2$

Result: Area of road widening for the 80 m stretch $\approx 285\ m^2$ (average of the two methods).

Precautions

- Offsets must be perpendicular to the chain line; keep the chain taut and straight.
- Do not exceed 15 m for perpendicular offsets by cross staff.

Viva Questions

1. What is an offset? Types (perpendicular, oblique).
2. Why is a chain survey suitable only for small, open areas?
3. Name the instruments to set perpendicular offsets.

Experiment 2: Distance between Two Inaccessible Points using Compass

Aim: To determine the distance between two inaccessible points A and B (e.g. across a river/pond) using a prismatic compass.

Apparatus: Prismatic compass, tripod, ranging rods, chain/tape, arrows, pegs.

Theory / Formulae

- Take a measurable baseline PQ and observe the bearings of A and B from P and Q.
- Angles are got from bearings: $\angle APQ$, $\angle BPQ$, $\angle AQP$, $\angle BQP$.
- Sine rule in triangles PAQ and PBQ: $PA = PQ \sin\beta_1 / \sin(\alpha_1 + \beta_1)$, $PB = PQ \sin\beta_2 / \sin(\alpha_2 + \beta_2)$
- Cosine rule in triangle APB: $AB^2 = PA^2 + PB^2 - 2\,PA \cdot PB \cos(\angle APB)$
- Check: back bearing = fore bearing ± 180°. Local attraction is checked by the difference of F.B. and B.B.

Procedure

1. Select stations P and Q such that A and B are visible from both; measure PQ accurately by chain.
2. Set up the compass at P, level, centre, and observe fore bearings of Q, A and B.
3. Shift to Q and observe bearings of P, A and B (check back bearing of PQ).
4. Compute the interior angles and solve the triangles for PA, PB and then AB.

Observation Table

Station	Sighted	F.B.	B.B.
P	Q	90°00′	—
P	A	30°00′	—
P	B	50°00′	—
Q	P	—	270°00′
Q	A	330°00′	—
Q	B	10°00′	—

Result: Distance AB computed from the sine and cosine rules (see sample in Experiment 7 for the same geometry).

Precautions

- Keep away from magnetic substances (steel, wires, vehicles).
- Read the bearing only after the needle comes to rest.

Viva Questions

1. What is local attraction and how is it corrected?
2. Difference between true and magnetic bearing.
3. What is magnetic declination?

Experiment 3: Plane Table Survey – Area of a Boundary by Radiation

Aim: To find the area of a given boundary by the radiation method of plane tabling.

Apparatus: Plane table with tripod, alidade, spirit level, plumbing fork with plumb bob, trough compass, drawing sheet, ranging rods, chain/tape, pins.

Theory / Formulae

- Plane table is set at one central station O; rays are drawn to all boundary stations A, B, C, D, E.
- Distances OA, OB… are measured and plotted to a convenient scale.
- Area = $\Sigma \frac{1}{2} r_1 r_2 \sin\theta$ of the triangles formed at O (or by planimeter/graph from the plotted figure).
- Three essentials of table setup: levelling, centring, orientation.

Procedure

1. Fix the sheet on the board, set up the table at O (central, visible to all stations), level it.
2. Mark the point o on the sheet over O with a plumbing fork; orient with the trough compass and draw the north line.
3. Sight each station with the alidade, draw rays, measure the distances and mark a, b, c, d, e to scale.
4. Join a-b-c-d-e-a to get the boundary plan; compute the area.

Observation Table

Ray	Station	Distance (m)	Scale plotted (cm)	Triangle
1	A	24.0	4.8	OAB
2	B	30.5	6.1	OBC
3	C	27.0	5.4	OCD
4	D	22.5	4.5	ODE
5	E	28.0	5.6	OEA

Result: Area of the boundary = ... m^2 (sum of the triangle areas).

Precautions

- Level the table properly and keep it undisturbed.
- Keep the alidade edge touching the pin while sighting.

Viva Questions

1. Name the methods of plane tabling.
2. What is orientation? Methods?
3. Advantages of plane table survey.

Experiment 4: Fly Levelling – Height of Instrument Method

Aim: To determine reduced levels of points using the height of instrument (collimation) method.

Apparatus: Dumpy/auto level, levelling staff, tripod, pegs, field book.

Theory / Formulae

- HI = RL of BM + BS; RL = HI − FS (or IS). After a change point: new HI = RL of change point + BS.
- Arithmetic check: $\Sigma BS - \Sigma FS$ = last RL − first RL.

Procedure

1. Set the level midway between BM and the first point; level using foot screws.
2. Take BS on the BM, then IS on intermediate points and FS on the change point.
3. Shift the instrument, take a BS on the change point and continue; close on a BM or the starting point.
4. Compute HI and RLs; apply the arithmetic check.

Observation Table

Station	BS	IS	FS	HI	RL	Remark
BM	1.525			101.525	100.000	BM
A		1.360			100.165	
TP1	2.115		0.985	102.655	100.540	CP
B		1.820			100.835	
TP2	1.980		1.435	103.200	101.220	CP
C			0.745		102.455	

Sample Calculation

$\Sigma BS = 1.525 + 2.115 + 1.980 = 5.620$; $\Sigma FS = 0.985 + 1.435 + 0.745 = 3.165$

$\Sigma BS - \Sigma FS = 2.455$; last RL − first RL = 102.455 − 100.000 = 2.455 ✓

Result: RL of C = 102.455 m; arithmetic check satisfied.

Precautions

- Hold the staff vertical (wave it); keep BS and FS distances equal.
- Do not shift the instrument between BS and FS.

Viva Questions

1. BS, IS, FS – define.
2. What is a change point?
3. Which method suits many intermediate points?

Experiment 5: Fly Levelling – Rise and Fall Method

Aim: To determine the reduced levels of points by the rise and fall method.

Apparatus: Level, levelling staff, tripod, pegs, field book.

Theory / Formulae

- Difference between consecutive staff readings: Rise = (previous reading) – (present reading) if positive, otherwise fall.
- RL = previous RL + rise (or – fall)
- Check: $\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall$ = last RL – first RL.

Procedure

1. Perform the field work as in Experiment 4.
2. Compute rise/fall between every two consecutive readings (the same staff reading pair, not across change point shift).
3. Find RLs and apply the three-fold check.

Observation Table

Station	BS	IS	FS	Rise	Fall	RL
BM	1.525					100.000
A		1.360		0.165		100.165
TP1	2.115		0.985	0.375		100.540
B		1.820		0.295		100.835
TP2	1.980		1.435	0.385		101.220
C			0.745	1.235		102.455

Sample Calculation

$\Sigma Rise = 0.165 + 0.375 + 0.295 + 0.385 + 1.235 = 2.455$; $\Sigma Fall = 0$

$\Sigma BS - \Sigma FS = 5.620 - 3.165 = 2.455$ = last RL – first RL ✓

Result: RL of C = 102.455 m (same as H.I. method).

Precautions

- Note readings to 5 mm/1 mm accuracy; avoid parallax.

Viva Questions

1. Compare HI and rise-and-fall methods.
2. What is parallax?
3. What is a bench mark? Types.

Experiment 6: Theodolite Survey – Horizontal and Vertical Angles by Repetition

Aim: To measure the horizontal angle between two points by the method of repetition and a vertical angle.

Apparatus: Vernier/digital theodolite, tripod, plumb bob, ranging rods, pegs.

Theory / Formulae

- Repetition reduces instrumental errors and reading errors: Angle = (final reading – initial reading)/n (number of repetitions).
- Face left and face right observations are averaged to eliminate collimation and horizontal axis errors.
- Vertical angle = mean of FL and FR vertical readings (after index correction).

Procedure

1. Centre, level and focus the theodolite at station O; set A and B as targets.
2. Set 0° on the plate, sight A (lower clamp), release the upper clamp and sight B; note the reading.
3. Release the lower clamp, sight A again, then upper clamp and sight B – this adds the angle; repeat n = 3 times; take the final reading.
4. Change face and repeat; take the mean.
5. For the vertical angle: sight the target, read the vertical circle on both faces.

Observation Table

Station	Face	Target	Initial	Final (after 3)	Angle
O	L	A,B	0°00′00″	135°20′40″	45°06′53.3″
O	R	A,B	0°00′00″	135°20′20″	45°06′46.7″

Sample Calculation

Horizontal angle (FL) = 135°20′40″/3 = 45°06′53.3″

Vertical angle: FL = 12°30′20″, FR = 12°30′40″ → mean = 12°30′30″

Result: Horizontal angle AOB ≈ 45°06′50″; vertical angle = 12°30′30″.

Precautions

- Centre and level accurately; do not touch the tripod while observing.
- Use the correct clamp and tangent screw sequence.

Viva Questions

1. Why is face change necessary?
2. What is least count of a theodolite?
3. Explain transiting.

Experiment 7: Theodolite Survey – Distance between Two Inaccessible Points

Aim: To find the distance between two inaccessible points A and B using a theodolite and a baseline.

Apparatus: Theodolite with tripod, ranging rods, tape, pegs.

Theory / Formulae

- Measure a baseline PQ. At P read $\alpha_1 = \angle QPA$ and $\alpha_2 = \angle QPB$; at Q read $\beta_1 = \angle PQA$ and $\beta_2 = \angle PQB$.
- $PA = PQ \sin\beta_1 / \sin(\alpha_1 + \beta_1)$; $PB = PQ \sin\beta_2 / \sin(\alpha_2 + \beta_2)$; $\angle APB = \alpha_1 - \alpha_2$
- $AB = \sqrt{(PA^2 + PB^2 - 2 \cdot PA \cdot PB \cos\angle APB)}$

Procedure

1. Select P and Q on level ground from where A and B are visible; measure PQ.
2. Set up at P: measure $\angle QPA$ and $\angle QPB$; set up at Q: measure $\angle PQA$ and $\angle PQB$ (both faces).
3. Compute PA and PB, then AB.

Observation Table

Station	Angle to A	Angle to B	Baseline
P	$\alpha 1 = 70°$	$\alpha 2 = 40°$	PQ = 50 m
Q	$\beta 1 = 60°$	$\beta 2 = 80°$	

Sample Calculation

$PA = 50 \sin 60° / \sin 130° = 56.53$ m; $PB = 50 \sin 80° / \sin 120° = 56.86$ m; $\angle APB = 30°$

$AB^2 = 56.53^2 + 56.86^2 - 2(56.53)(56.86)\cos 30° \rightarrow AB = 29.35$ m

Result: Distance AB = 29.35 m (for the sample data).

Precautions

- Measure the baseline carefully; choose well-conditioned triangles (angles between 30° and 120°).

Viva Questions

1. What is a well-conditioned triangle?
2. Which angles are measured in this method?
3. Source of errors?

Experiment 8: Theodolite Survey – Height of a Far Object

Aim: To determine the height of an object whose base is inaccessible.

Apparatus: Theodolite, tripod, levelling staff, tape, ranging rods.

Theory / Formulae

- Two instrument stations P and Q in line with the object, distance b apart (Q farther). Angles of elevation α_1 (at P) and α_2 (at Q).
- $h = b \tan\alpha_1 \tan\alpha_2 / (\tan\alpha_1 - \tan\alpha_2)$ (above the instrument axis); total height $H = h + h_i$ (instrument height at P).
- Distance from P to the foot of the object: $D = h / \tan\alpha_1$

Procedure

1. Set up the theodolite at P, level it, measure the height of the instrument axis.
2. Measure the angle of elevation α_1 to the top of the object (both faces).
3. Move to Q in the same vertical line, measure $b = PQ$ and the angle α_2; compute h.

Observation Table

Station	Angle of elevation	Instrument height	b
P	$\alpha 1 = 40^\circ$	1.45 m	30 m
Q	$\alpha 2 = 25^\circ$	1.45 m	

Sample Calculation

$h = 30 \times \tan 40^\circ \times \tan 25^\circ / (\tan 40^\circ - \tan 25^\circ) = 30 \times 0.8391 \times 0.4663 / 0.3728 = 31.49$ m

Height of object $\approx 31.49 + 1.45 = 32.94$ m; $D = 37.53$ m

Result: Height of the object = 32.94 m (sample data).

Precautions

- Keep P, Q and the object in one vertical plane; measure angles on both faces.

Viva Questions

1. When is the single-plane method used?
2. What is angle of elevation/depression?
3. What is the instrument axis?

Experiment 9: Total Station – Area and Perimeter

Aim: To determine the area and perimeter of a plot using a total station.

Apparatus: Total station, tripod, reflecting prism with pole, tape, field book.

Theory / Formulae

- Total station measures horizontal angle, vertical angle and slope distance electronically and computes coordinates (N, E, RL).
- Perimeter = $\Sigma \sqrt{(\Delta E^2 + \Delta N^2)}$
- Area by the coordinate formula: $A = \frac{1}{2} |\Sigma (x_i y_{i+1} - x_{i+1} y_i)|$ (or use the built-in Area program).

Procedure

1. Set up, centre and level the instrument at a station; enter the job name, station coordinates, instrument/prism heights and atmospheric corrections.
2. Back-sight a known point and set the orientation (HA = 0° or known azimuth).
3. Sight the prism at each boundary corner in order and record the points.
4. Use the Area function / download data; compute area and perimeter.

Observation Table

Point	E (m)	N (m)
A	0	0
B	60	0
C	80	45
D	30	70
E	-10	40

Sample Calculation

$A = \frac{1}{2} |\Sigma (x_i y_{i+1} - x_{i+1} y_i)| = 4425.0\ m^2$

Perimeter = AB + BC + CD + DE + EA = 256.38 m

Result: Area = 4425.0 m^2, perimeter = 256.38 m (sample coordinates).

Precautions

- Level the instrument; keep the prism pole vertical; protect the instrument from rain/sun.

Viva Questions

1. What does EDM mean?
2. Explain back sight and orientation.
3. Advantages of a total station over a theodolite.

Experiment 10: Total Station – Distance between Two Inaccessible Points

Aim: To determine the distance between two inaccessible points using the Missing Line Measurement (MLM) function of a total station.

Apparatus: Total station, tripod, prisms with poles.

Theory / Formulae

- From the instrument station O, the coordinates of A and B are obtained from measured HA, VA and SD.
- Horizontal distance $HD = SD \cos V$ (V = elevation angle); $E = E_0 + HD \sin Az$; $N = N_0 + HD \cos Az$
- $AB = \sqrt{[(E_B - E_A)^2 + (N_B - N_A)^2]}$; with RL, slope distance $= \sqrt{[HD^2 + (\Delta RL)^2]}$

Procedure

1. Set up the instrument at O visible to both A and B; enter station data.
2. Sight the reflector (or reflectorless target) at A and measure; then at B.
3. Use MLM (Menu → Programs → MLM) to display the distance between A and B directly; note HD, slope distance and ΔH.
4. Verify by calculation from the coordinates.

Observation Table

Point	HA	SD (m)	V angle	HD (m)
A	30°00'00"	42.50	88°00'	42.48
B	110°00'00"	38.20	90°30'	38.20

Sample Calculation

A: $E = 42.48 \sin 30^\circ = 21.24$, $N = 42.48 \cos 30^\circ = 36.79$; B: $E = 38.20 \sin 110^\circ = 35.90$, $N = 38.20 \cos 110^\circ = -13.07$

$AB = \sqrt{[(35.90 - 21.24)^2 + (-13.07 - 36.79)^2]} = 51.97$ m

Result: Horizontal distance AB computed (≈ 52.0 m for the sample data); compare with the MLM display.

Precautions

- Check the prism constant and atmospheric settings.

Viva Questions

1. What is MLM?
2. Slope distance vs horizontal distance.
3. What is a prism constant?

Experiment 11: Setting Out a Curve

Aim: To set out a simple circular curve by the deflection angle (Rankine's) method.

Apparatus: Theodolite, chain/tape, ranging rods, arrows, pegs.

Theory / Formulae

- Tangent length $T = R \tan(\Delta/2)$; curve length $L = \pi R\Delta/180$; long chord $= 2R \sin(\Delta/2)$
- T_1 chainage = chainage of PI − T; T_2 chainage = $T_1 + L$
- Deflection angle for a chord c: $\delta = (c/2R)$ radians $= 1718.9\ c/R$ minutes; deflection angles are cumulative; the last one must equal $\Delta/2$ (check).

Procedure

1. Locate the PI, measure Δ; compute T, L and the chainages of T_1 and T_2; fix T_1 and T_2 on the tangents.
2. Set the theodolite at T_1 with $0°$ on the line to PI.
3. Set the first deflection angle and measure the first sub-chord from T_1 to get peg 1; then set the cumulative angle and swing a full chord from peg 1 with the tape to intersect the line of sight to get peg 2.
4. Continue to T_2 and check that the final deflection equals $\Delta/2$.

Observation Table

Peg	Chainage (m)	Chord (m)	δ (deg)	Cumulative δ
T1	884.53	—	0	0°00′
1	900.00	15.47	2.216	2°13.0′
2	920.00	20.00	2.865	5°4.8′
3	940.00	20.00	2.865	7°56.7′
4	960.00	20.00	2.865	10°48.6′
5	980.00	20.00	2.865	13°40.5′
6	1000.00	20.00	2.865	16°32.4′
7	1020.00	20.00	2.865	19°24.3′
8	1040.00	20.00	2.865	22°16.2′
9	1060.00	20.00	2.865	25°8.1′
10	1080.00	20.00	2.865	27°59.9′
T2	1093.97	13.97	2.001	30°0.0′

Sample Calculation

$R = 200$ m, $\Delta = 60°$, PI chainage = 1000 m

$T = 200 \tan 30° = 115.47$ m; $L = \pi \times 200 \times 60/180 = 209.44$ m; $T1 = 884.53$ m; $T2 = 1093.97$ m

Result: The curve is set out with pegs at 20 m chainage; final cumulative deflection = 30.0° (check).

Precautions

- Measure chords with the tape tight and horizontal; check closure at T2.

Viva Questions

1. Define deflection angle.
2. Types of curves (simple, compound, reverse, transition).
3. What is a sub-chord?

Experiment 12: Determining the Levels of Contours

Aim: To determine the levels of contours of an area by the grid (squares) method of levelling.

Apparatus: Level, levelling staff, tape, ranging rods, pegs.

Theory / Formulae

- A contour is a line joining points of equal elevation; contour interval (CI) = vertical distance between consecutive contours.
- Divide the area into squares (grid) of 10–20 m; find the RL at every grid corner by levelling.
- Interpolation: position of a contour between two points of RL H_1 and H_2 at spacing d from the first: $x = d\,(H_c - H_1)/(H_2 - H_1)$
- Contours never cross (except at overhanging cliffs/caves); closely spaced = steep slope.

Procedure

1. Lay out a grid over the area and mark the corners with pegs.
2. Set up the level, take a BS on a BM and note IS at each grid corner (HI method); close on the BM.
3. Compute RLs of all corners; choose the CI (say 0.5 m or 1 m).
4. Interpolate the contour positions between grid points and join points of equal RL smoothly.

Observation Table

Grid pt	A1	A2	A3	B1	B2	B3
RL (m)	100.40	101.30	102.10	100.80	101.60	102.50

Sample Calculation

Contour 101.00 between A1 (100.40) and A2 (101.30), d = 20 m: $x = 20 \times (101.00 - 100.40)/(101.30 - 100.40) = 13.33$ m from A1

Contour 102.00 between A2 and A3: $x = 20 \times (102.00 - 101.30)/(102.10 - 101.30) = 17.5$ m from A2

Result: Contour plan prepared at the chosen contour interval; the contour lines are drawn on the plotted grid.

Precautions

- Close the levelling circuit; contour lines must be smooth curves.

Viva Questions

1. Define contour interval and horizontal equivalent.
2. Characteristics of contours.
3. Uses of contour maps.